

1. Hardware & Interface Overview

The device is controlled via a centralized interface consisting of two primary input/output components:

- **Rotary Encoder Knob with Push-button:** Used for navigating menus, selecting modes, and fine-tuning parameters (such as speed, sensitivity, and durations).
- **24-LED RGB Ring:** Provides real-time visual feedback regarding the current operating mode, motor speed, and pressure sensor telemetry at a 60Hz update rate.

Audio Feedback: The vibrator motor is utilized as a rudimentary speaker. It will emit distinct tones to signal power-on, power-off, and critical hardware warnings (such as the pressure sensor exceeding readable limits).

2. Power and Navigation Control

The push-button integrated into the encoder knob dictates the state machine of the device. The system distinguishes between three types of button presses based on the duration the button is held down.

Press Type	Duration	Primary Function
Short Press	< 600 ms	Cycles forward through the main operating modes or configuration menus.
Long Press	600 ms - 2500 ms	Jumps from a main operating mode into the associated Configuration Menu, or exits a Configuration Menu back to Manual mode.
Very Long Press	> 2500 ms	Suspends device operation (Standby/Off). The LEDs and motor turn off, and a descending audio tone is played. The device remains off until the button is pressed again.

Note on Data Retention: User preferences, including maximum speed, sensitivity, target edges, and release duration, are persistently stored in the microcontroller's EEPROM. These settings are saved automatically when exiting the respective configuration menus and will survive a complete power cycle.

3. Main Operating Modes

By utilizing a **Short Press**, the user cycles sequentially through the four primary operating modes. The current mode is easily identified by the color of the LED cursor on the ring.

3.1 Manual Control Mode (Red Cursor)

Indicator: Red LED cursor. Background LEDs display current pressure changes via a Green/Yellow/Red gradient.

Function: Direct, manual control over the vibrator's motor speed.

Use Case: Standard vibrator operation or baseline testing.

Knob Action: Turning the knob scales the motor PWM output linearly from 0 to the absolute maximum (255).

3.2 Abrupt Automatic Edging Mode (Blue Cursor)

Indicator: Blue/Purple LED cursor.

Function: The original automatic edging algorithm. The motor linearly ramps up to the configured Max Speed over 30 seconds. If a sudden spike in pressure (indicating a near-orgasm clench) exceeds the defined limit, the motor shuts off abruptly. After a 15-second cool-down, the ramp-up restarts.

Use Case: Strict, highly distinct start/stop edging

Knob Action: Adjusts the orgasm detection sensitivity. Turning the knob changes the acceptable pressure delta (limit) before a cutoff is triggered.

3.3 Smooth Automatic Edging Mode (Cyan Cursor)

Indicator: Cyan/Teal LED cursor.

Function: A proportional edging algorithm. Instead of cutting power instantly, the controller continuously calculates how close the current pressure delta is to the user-defined limit. As the pressure approaches the limit, the motor speed is dynamically and proportionally throttled down. If the limit is fully breached, the motor drops to zero.

Use Case: Fluid, continuous edging where the vibration softens organically as arousal peaks, allowing the user to "ride the edge" without abrupt stops.

Knob Action: Adjusts the orgasm detection sensitivity limit.

3.4 Auto-Release Mode (**Magenta Cursor**)

Indicator: Magenta/Hot Pink LED cursor.

Function: An advanced edging sequence manager. This mode utilizes the smooth proportional throttling logic from the Cyan mode, but actively counts the number of times the user crosses the pressure limit (an "edge"). Once the user hits the pre-configured target number of edges, the device initiates a "Release Phase." During the Release Phase, the LEDs flash white, and the motor runs continuously at the maximum configured speed for a predetermined duration to facilitate an orgasm. Afterwards, the counter resets.

Use Case: Goal-oriented edging sessions requiring a strict, pre-defined number of peaks before a forced release is granted. Define release duration (duration of full power vibrator after release is allowed and detected) to facilitate either a ruined orgasm (short release duration, ~2 seconds), a normal, full orgasm (normal release duration, 5-20 seconds) or post orgasm torture (long release duration, >20 seconds).

Knob Action: Adjusts the orgasm detection sensitivity limit.

Live Progress Tracking: The Auto-Release mode features a dedicated visual interface mapped to the LED ring to track the session. The final position on the ring is illuminated with a fixed gold/yellow LED, representing the baseline target. As the user achieves valid edges, a pink LED cursor advances clockwise toward the gold target, providing progress feedback.

Edge Cooldown: To prevent a single sustained clench, rapid internal pressure fluctuations or deliberate manipulation from artificially inflating the edge count, the software enforces a strict 15-second debounce window. An edge is only successfully counted—and the pink cursor advanced—if a minimum of 15 seconds has elapsed since the last recorded edge. This is independent from the vibration ramping up again. If you cross the pressure limit during the 15-second cooldown window, the device will still drop the motor speed to zero to provide the necessary physical cooldown, but the edge counter will ignore the event.

4. Configuration Menus

Configuration menus are accessed via a **Long Press** from the main operating modes.

4.1 Maximum Motor Speed (Green Menu)

Access: Long press from Manual, Abrupt Auto, or Smooth Auto modes.

Indicator: Green LED background animation that visually represents a speed ramp-up.

Function: Sets the absolute maximum speed ceiling that the automatic modes are allowed to reach during their 30-second ramp-up phases.

Knob Action: Adjusts the max speed ceiling.

4.2 Ramp-Up Time Setting (Solid Orange Menu)

Access: Reached via a Short Press while currently inside the Maximum Motor Speed (Green) menu.

Indicator: Solid Orange background LEDs.

Function: Defines the precise duration it takes for the vibrator motor to linearly accelerate from a stopped state up to the ceiling defined in the Maximum Motor Speed menu. This setting fundamentally dictates the pacing and urgency of the edging cycle across all three automatic modes (Abrupt Auto, Smooth Auto, and Auto-Release).

Knob Action: Adjusts the ramp-up duration. Turning the encoder maps the 24 LEDs to a time range spanning from 5 seconds to 120 seconds, moving in 5-second increments.

Exit: A Short Press saves the newly selected ramp time to the EEPROM memory and transitions the user into the Pressure Debug Monitor (White) menu.

A Long Press immediately saves the setting to memory and exits the configuration loop entirely, returning the device to Manual Mode (Red).

4.3 Target Edges Setting (Solid Magenta Menu)

Access: Long press exclusively from the **Auto-Release** mode.

Indicator: Solid Magenta background LEDs.

Function: Defines how many edges the user must achieve in Auto-Release mode before the final release vibration is triggered.

Knob Action: Selects the target edge count, ranging from 2 to 48 edges in increments of 2.

4.4 Target Edge Variance Setting (Solid Cyan Menu)

Access: Reached via a Short Press while currently inside the Target Edges (Solid Magenta) menu.

Indicator: Cyan background LEDs, blocks of 4-5.

Function: Introduces a dynamic "roulette" element to the Auto-Release mode to prevent extreme predictability. This menu dictates a randomized +/- variance that is secretly applied to the baseline target edge count at the start of every release cycle. For example, if the baseline target is set to 4, and the variance is set to 2, the device will secretly require anywhere between 2 and 6 edges before triggering the release.

If activated, the pink progress cursor in Auto-Release mode intentionally lies to the user. It only displays progress relative to the baseline target. If the randomized variance selects a target higher than the baseline, the Pink cursor will reach the Gold LED and stop. The user will be held in "overtime," forced to continue edging without visual feedback until the hidden target is finally met.

Note: The software prevents the randomized target from ever dropping below 1. A release phase cannot be triggered instantly upon starting a session. Please take care to set a realistic variance depending on the set number of target edges for an enjoyable experience.

Knob Action: Adjusts the variance range. Turning the encoder maps the variance from 0 to 6, visually indicated by the position of the White cursor over the Cyan LEDs. Setting this to 0 completely disables the roulette feature, ensuring the release triggers exactly on the baseline target every time.

Exit: A Short Press saves the variance setting to persistent EEPROM memory and transitions into the Release Duration (Yellow) menu.

A Long Press immediately saves the setting and exits the configuration loop, returning the device to Manual Mode (Red).

4.5 Release Duration Setting (Solid Yellow Menu)

Access: Short press from the Target Edges menu.

Indicator: Solid Yellow background LEDs.

Function: Defines the length of the "Release Phase" in Auto-Release mode.

Knob Action: Selects the duration of the final maximum-speed vibration, ranging from 2 to 48 seconds in increments of 2.

Exit: A Short press from this menu transitions into the standard Max Speed (Green) menu, or a Long press saves and exits to Manual mode.

4.6 Pressure Debug Monitor (White Menu)

Access: Short press from the Max Speed (Green) menu.

Indicator: Single White LED cursor.

Function: A diagnostic tool that maps the raw analog voltage from the pressure sensor directly to the LED ring. The motor is forced off in this mode.

Use Case: Verifying that the inflatable plug is registering pressure correctly or troubleshooting hardware sensor sensitivity.

5. Technical Safety & Troubleshooting

- **Pressure Sensor Railing:** If the internal analog pressure reading exceeds a value of 4030, the system detects a sensor overload. The motor will emit three high-pitched warning beeps. If left unchecked, this condition leads to missed edge detections. To resolve this, adjust the hardware trimpot on the circuit board to lower the baseline analog gain.
- **Hardware Diagnostics:** The device streams real-time telemetry (Target Motor Speed, Raw Pressure, and 25-Second Running Average Pressure) over the USB Serial interface at a baud rate of 115200. This data can be monitored via the Arduino IDE Serial Monitor for advanced fine-tuning.
- **Thermal Management:** The software limits the overall LED brightness to a value of 50 to restrict current draw. Modifying the source code to display all LEDs at maximum white brightness (RGB 255,255,255) simultaneously is strictly prohibited, as it may overheat the voltage regulator and cause unexpected microcontroller resets.

Changelog:

20260721.2 – Initial release, smooth edging and auto-release mode implemented.

20260721.3 – Ramp-up time setting added.

20260721.4 – Edge progress indicator for auto-release mode implemented. Random (configurable) variance added for target edges. Timeout for edge progress counter implemented to prevent rapid triggering and increase in edge count.

20260721.5 – Standby mode fixed.

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